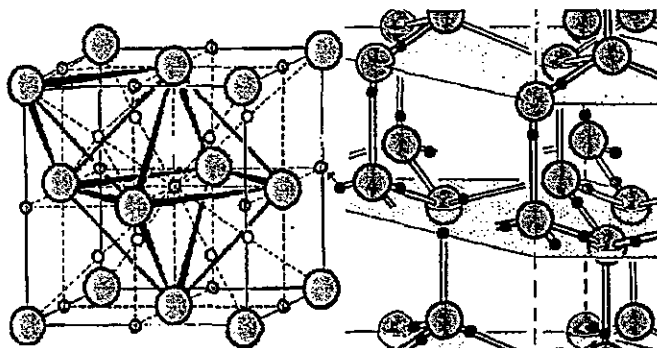


TO COLDLY GO WHERE NO ONE HAS GONE BEFORE

Palladium (at left) is shown loaded with an isotope of hydrogen. Water (on the right) is the source of the isotopic fuel for cold fusion. All of the sites of possible loading are shown as "filled" in this figure.



A BRIEF REVIEW OF THE HISTORY OF COLD FUSION

by Dr. Eugene F. Mallove

It turns out that the original discoverers, Drs. Pons and Fleischmann, were correct in their claims of power production from water and these have now been verified in various aspects in hundreds of laboratories around the world, including excellent labs in the United States. Pons and Fleischmann are now working on cold fusion with full support by a major Japanese corporation (IMRA Europe) near Nice, France. Of course, after the cold fusion story's initial huge publicity in the spring of 1989, the whole matter seemed to cool off and disappear. Supposedly "good" universities and laboratories "proved" that Pons and Fleischmann were buffoons or frauds. Glad to say, this common perception about cold fusion is 100% wrong.

You can begin to verify every assertion that I am making by examining the enclosed material, but let me summarize the major points:

Continued on page 4

**STATEMENT BY THE HONORABLE
DICK SWETT BEFORE THE SCIENCE,
SPACE & TECHNOLOGY
COMMITTEE, ENERGY
SUBCOMMITTEE HEARING:
"INVESTING IN TECHNOLOGIES FOR
AMERICA'S ENERGY FUTURE"**

WASHINGTON February 16, 1993 ---

"I am pleased to see our subcommittee holding a hearing on an issue of such critical importance. Last year's energy bill did much to help move our country in the right direction,

Continued on page 2

In March 1989 electrochemically induced nuclear fusion reactions were reported, but were initially very difficult to reproduce. By 1993 many experiments have demonstrated that anomalous nuclear processes occur in palladium which has been highly loaded with deuterium.

COLD FUSION TIMES is devoted to novel research and developments in the field popularly known as "cold fusion". The field has expanded considerably since 1989. So this newsletter will report to its readers on the current state-of-the-art.

This year, on January 16, at the MIT IAP all-day seminar on cold fusion Prof. Peter L. Hagelstein gave a fantastic detailed presentation on this subject leaving the participants in the auditorium in rapt attention.

NEWS FROM JAPAN:

SUMMARY OF THE THIRD INTERNATIONAL CONFERENCE ON COLD FUSION

By: Peter L. Hagelstein
Massachusetts Institute of Technology
Research Laboratory of Electronics
Cambridge, Massachusetts

We review highlights of the international cold fusion conference that was held recently in Nagoya, Japan. Excess heat results in heavy water electrolysis experiments constitute the observations with the most important potential applications. Experiments in gas phase systems exhibit fast particle and gamma emission that make progress toward elucidating mechanisms. The evidence in support of a light water heat effect has improved.

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COLD FUSION TIMES

volume 1 number 1
SPRING 1993

A NEWSLETTER
DEVOTED TO
RELEVANT RESEARCH
and DEVELOPMENTS
OF SAFE, RELIABLE
POWER SOURCES

"It turns out that the
original discoverers,
Drs. Pons and
Fleischmann, were
correct"

Eugene F. Mallove, ScD

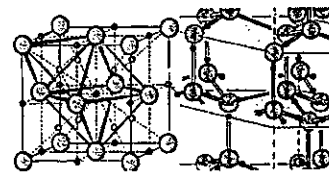
"Over 300 attendees
participated, listening to
about 27 oral
presentations and
looking over roughly 80
poster papers."

Prof. Peter Hagelstein

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JET Technology

COLD FUSION TIMES



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STATEMENT BY CONG. SWETT

but I believe that there is much more that we must do to accelerate our nation's transition from being a fossil-fuel based economy to being an energy self-sufficient economy based on alternate and renewable energy sources. I'm sure our panelists will have much to say about new energy technologies, and I expect that I will have numerous questions for them. I am particularly interested, however, in a field which is commonly referred to as cold fusion.

Back in 1989, cold fusion first hit the media when Pons and Fleischmann made their startling announcement. Their announcement, however, turned out to be premature. The DOE issued a negative report, and cold fusion research has been languishing, at least in this country, ever since.

Despite unfortunate events of 1989, cold fusion research has continued. Japan has launched a substantial research effort, and, in this country, the Electric Power Research Institute (represented on today's panel), has undertaken a \$12 million cold fusion research effort. Scientists at SRI and Los Alamos National Laboratory, along with scientists at various other places around the country and around the world, report getting excess heat from the electrolysis of "heavy" water. I have recently received letters about cold fusion which I would like to have entered into the record. One is from Dr. Edmund Storms, a scientist at Los Alamos. Another is from a constituent of mine, Dr. Eugene Mallove, author of "Fire from Ice: Searching for the Truth Behind the Cold Fusion Furor." Both letters address the need for increased government attention to cold fusion.

Not being a scientist, I'm not in a position to evaluate the technical merit of recent cold fusion research. It does seem to me, however, that there exists sufficient factual evidence to warrant another comprehensive scientific review of cold fusion.

In any case, I hope that our subcommittee will play a role in helping to find the truth behind the cold fusion furor. The mere possibility that there may exist a new source of energy that uses water as a fuel makes it too important to pass up."

Ed. note: We salute Congressman Swett for his insight, courage, and dedication

CONTINUED FROM PAGE 1:
SUMMARY BY HAGELSTEIN

INTRODUCTION

The Third International Cold Fusion Conference took place in Nagoya, Japan between October 21 and October 25, 1992. Over 300 attendees participated, listening to about 27 oral presentations and looking over roughly 80 poster papers.

The results presented at this conference were overall technically much stronger than last year's conference, and benefitted by a very strong showing from the Japanese contingent. I will first itemize what I thought were some of the most interesting new experimental results presented at the conference.

I admit to having numerous biases. One bias is that I believe the observations of excess power are ultimately the most important, both scientifically and technologically. Another bias is that I favor results which in my view help to elucidate reaction mechanisms.

Following the discussion of significant positive results, I review abstracts and presentations of negative results. Coming from the theory end of the field, I felt that it was appropriate for me to survey the theory papers which were presented (in the following section); in this case, it was possible to include a larger fraction of the papers submitted. Having my own theory as to the origin of the effect, I warn the reader that my discussion of theory necessarily carries a bias in favor of my world view; it is my hope that this discussion will be useful in spite of this bias. Almost as interesting in some cases as what was presented, was what was not presented; a discussion of work that was absent is presented before the summary and conclusions.

For a review as long as this one, there are many issues and many details, most of which I have made a serious attempt to get right. I would hope that I will not make enemies of those whose work I did not include (which at this point will include about half of all papers submitted). This review was constructed from preprints, notes, memory, and discussions with many people in the field -- should the reader note errors or misconceptions, I would appreciate corrections.

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C.F. CONFERENCE AT MIT

(Excerpted from Fusion Facts January 1993 -
Editor in Chief Hal Fox)

Cambridge (1/16/93) - Dr. Mitchell Swartz organized a cold fusion conference which was held during the Independent Activities Period at the Massachusetts Institute of Technology (Cambridge, MA) on Saturday, January 16, 1993. The participants at the conference included the following speakers and varied subjects.

Dr. Swartz showed video films of some of the more recent Pons-Fleischmann work in France, tapes which they graciously provided for the seminar. These impressive videos were similar to those shown at the Nagoya conference in October, and were augmented by other video tapes.

Eugene Mallove (author of Fire From Ice, the best book to date on cold fusion) discussed the history of cold fusion.

Vesselin Nominiski (a Bulgarian scientist who with his famous father were the 5th group to replicate the Pons-Fleischmann Effect) discussed the problems that Cal Tech and MIT had with calorimetry in their early cold fusion work.

Peter Hagelstein (Associate Professor, MIT) summarized the latest experimental evidence from the Nagoya conference and discussed his theory of cold fusion.

Robert H. Rines (Chairman of law school at Franklin Pierce Law Center) discussed the current patent problems and caused by the patent office's treatment of cold fusion as not being a real technology.

Larry Forsley (from Rochester, N.Y.) made a brief presentation on his recent cold fusion experiment in which the palladium cathode boiled away the heavy water electrolyte.

Hal Fox (Editor of Fusion Facts) presented a paper on the commercialization of various types of cold fusion phenomena.

Dr. Swartz and Richard Shyduroff (Director of the MIT Entrepreneurs Club) were scheduled to speak on isotope loading and possible commercial enterprises, respectively, but were unable to give their talks due to the interesting and unexpected lengthy presentations of two of the speakers. [The entire program lasted eight hours (almost two hours beyond the scheduled conclusion!).]

The conference was attended by about 90 persons including faculty, students, and the public.

The presentation by Dr. Hagelstein made no apologies for the new science of cold fusion. He presented some of the latest data that was presented at the Nagoya conference; stated his acceptance of the reality of cold fusion; and presented a

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"There exists
sufficient factual
evidence to warrant
another
comprehensive
scientific review of
cold fusion"

HONORABLE DICK SWETT
FEBRUARY 16, 1993

"Pons and Fleischmann
announced ... excess power
production (on the order of
1000 Watts/cm{3})
corresponding to a factor of
10 power gain, and that they
had done so 11 times."

summary of his theory of how cold fusion can be explained from
fundamental physical principles.

Robert H. Rines discussed the problems with the current patent office
treatment of cold fusion patents and made the following suggestions: Do
not use the term cold fusion in any patent applications. If the application is
rejected then the applicant has the option of filing a continuation in
addition to a possible appeal of the rejection. If you have a successful
experimental device or process have your work replicated by a party not
involved with your patent application and have each person write a
testimonial letter stating that your process works and he has replicated it.
He reminded that inventors need not be concerned if their patent is not
issued soon as it may run out before cold fusion has commercial value.

This was followed by Hal Fox's presentation, which was clearly much
more optimistic about the earlier and wider commercialization of cold
fusion. The main thrust of Hal Fox's presentation was that there are now
several devices in addition to the original Pons-Fleischmann cell by which
nuclear reactions can produce excess heat. Hal suggested that the
audience save some of their healthy skepticism (or dedicated denial) for the
following methods of producing excess heat: light-water, nickel-cathode
electrochemical cells; molten-salt electrochemistry cells; gas-loaded
devices; and capillary fusion. Commercialization potential was defined as
the capability of a device to produce over 300% excess heat as compared to
the input power.

Eugene Mallove presented a lengthy review of the history and politics of
cold fusion. He began with the idea that nature was atomic (about 400
B.C.). Mallove continues to be one of the most vocal supporters of the
concept that the government has an right to deny the existence of cold
fusion. He has been very active in bringing the reality of cold fusion to the
attention of government officials (both appointed and elected). Mallove
planned a second presentation showing that cold fusion had the potential
for the future propulsion of space vehicles, but the seminar ran out of time
on this topic.

Larry Forsley showed the cell structure that was involved in the January 4,
1993 "turn on" of the palladium cathode that resulted in the "boil-off" of
the heavy-water electrolyte. Forsley claims that he believes he has a
repeatable process for loading and that he obtained large amounts of excess
heat from a Pons-Fleischmann cell. A CCD camera and large-screen
monitor was provided so that the entire audience could examine the actual
cracked calorimeter in-situ.

At the end of the conference, Dr. Swartz initiated discussions for the
planning of the first "HEAT-OFF". This will be a contest between small
(non-industrial size) cold fusion units. JET Technology began the
planning. FEAT has now joined the growing list of sponsors. This
addition of FEAT to the sponsors was announced at the seminar when Hal
Fox offered to donate a palladium model for the winner of the "HEAT-OFF" contest.

Because of the rapidly developing technologies in cold fusion
research, COLD FUSION TIMES is available for home or office
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COLD FUSION TIMES will attempt to bring up-to-date and
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COLD FUSION TIMES

LESSONS OF HISTORY

by Jed Rothwell

1. New ideas & theories are usually rejected out of hand. This is true of correct ideas, incorrect ones, big ones, and little ones, including many that do not seem controversial now. A century ago, when someone suggested that the newly invented "zipper" be used to close the fly on men's trousers, the idea was met with gales of laughter and derision. When I suggested in 1979 that the newly invented microprocessor might eventually take away minicomputer business, the management of my company (and of IBM, and DEC) thought the idea was absurd. The acceptance -- or rejection -- of an idea cannot be used to predict anything.

2. Experimental evidence, no matter how simple and overwhelmingly convincing, is usually ignored for a period of years. You can fly an airplane in front of hundreds of people, you can circulate photographs and data, but people will go on denying that airplanes exist. You can string electric lights on poles, but experts will not bother coming out to check, even if you are the most important inventor in the world, and you have backing from J.P. Morgan. You can show a video of a boiling cell, and hand out data, but the "skeptics" will ignore you, even if Toyota has given you carte blanche support.

3. Popular support, and the opinion of experts, is no guide to anything. Never listen to experts who have a vested interest in the old ways of doing things. Only people who have actually spent months of hard work performing the experiment for themselves have any authority.

4. In science, people's opinions count for nothing. Previously established theories count for nothing. Only one thing matters: replicated experimental evidence. When sigma 90 results from 200 superb experiments say there are megajoules per mole of heat, then any theory or previous knowledge that would deny that fact is wrong.

5. People often succumb to the illusion that we have reached "the end of history" (F. Fukuyama's term). They think that all is known, and all knowledge is complete and perfect. Skeptics think that existing theories are perfect and can never be overturned or changed.

When something like CF comes along, it terrifies such people. They were too frightened and outraged to look through Galileo's telescope, and they are terrified of McKubre's data, because it shows that the world is ever-changing, and there will always be more in heaven and earth than dreamt of in our philosophy. These conservatives crave continuity, predictability and control.

CONTINUED FROM PAGE 2 :
SUMMARY BY HAGELSTEIN

SURVEY OF POSITIVE RESULTS

In the absence of current flow, film-boiling limits the heat flow from the cathode at cathode temperatures higher than about 120 Deg. C; the maximum heat flux from the rod under these conditions is limited to I think somewhere near 125 Watts/cm². The surface area of the cathode, including the top and bottom, is 0.85 cm², which leads to an observed average heat flux of about 170 Watts/cm². This number is comparable to, but greater than my version of the film-boiling limit given above, and was a potential cause for concern.

Pons and Fleischmann have considered this effect, and have found experimentally that the presence of current flow delays the onset of film-boiling to higher temperatures and higher heat fluxes. In their conference proceeding, they claim⁽²⁾ to have observed heat transfer rates during electrolysis in separate experiments which are between 1-10 kW/cm². I consider this result to be very important.

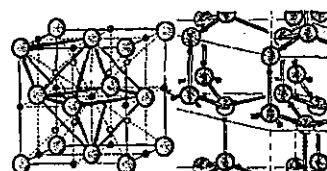
The cathode gets very hot in these experiments. Pons and Fleischmann have observed the Kel-F supports at the base of the cathodes to melt, from which the presence of temperatures in excess of 300 Deg. C are inferred. A direct measurement of the cathode temperature is currently problematic; Pons is currently interested in practical proposals as to how to do this without impacting the electrochemistry.

A common misunderstanding often occurs in the discussion of the results of Pons-Fleischmann experiments which is of interest here. It is sometimes argued that the energy production during a short event can be disregarded, since there may exist energy storage mechanisms which could have been collecting energy at a low level for a long period of time. For example, the total energy output from this experiment would not be very much larger than the total input energy if no heat excess had occurred prior to the boiling event (1 watt-day = 86.4 kJ). This type of argument seeks to make palatable the notion that since the total energy excess measured over days is small compared to the input (and hence there might exist a signal to noise problem in the measurement), the measurement can be dismissed. As discussed above, this type of argument completely misses a key implication of the experiment -- specifically, that there exists no known physical mechanism which could store the energy observed to be released during the boiling episode.

It is true of this experiment as well as of others to be described below, that no products or "ashes" of the heat have been found and verified that are commensurate with the energy production. This will be discussed further below.

2. M. McKubre^[3,4] described experiments done at SRI during the past several years. They have developed closed cell flow calorimeters, which are ideally *first principle* calorimeters (which means that the heat flow out of the cell goes into the flowing water coolant, and the power generation is determined by measuring the mass flow rate and output to input flow temperature difference, with no calibration required). They have succeeded in reducing the conduction losses (which are not first principle contributions) down to the order of one per cent, and then they calibrate the Fick's law constant associated with the conduction losses. Overall, the SRI calorimeters achieve a relative accuracy in the calorimetric measurements which is on the order of a few tenths of a per cent.

The SRI group reported the development of a procedure at Como that appeared to yield excess heat essentially every time, and this method is described in the Como conference proceedings. Highlights of these experiments were discussed. A significant



"Thanks are in order especially to Peter Hagelstein for this paper, but also to Jed Rothwell (and daughter Naomi) for working it into a format consistent with facile presentation."

"Prof. Hagelstein sent me this document in "Latex" format. I converted it to Word Perfect, and then to 80 column ASCII, with the following conventions:

- Superscripts, including footnote numbers, are shown between curly brackets: {4}He
- Subscripts are shown between square brackets: D[2]O
- Italics are marked by asterisks: *ab initio*
- Greek letters and symbols are spelled out: sigma, mu, Deg."

Jed Rothwell

COLD FUSION TIMES

CONTINUED FROM PAGE 1 :
HISTORY BY MALLOVE

1. Cold fusion (whatever its detailed scientific explanation turns out to be) has been demonstrated to be real and a spectacular source of energy from water. It is a new form of virtually radiationless nuclear energy. The implications of this are mind-boggling, needless to say.

2. Cold fusion is being aggressively developed by at least 15 major Japanese corporations in conjunction with Japan's Ministry of International Trade and Industry (MITI).

3. The U.S. Department of Energy convened a panel in mid-1989 that dismissed cold fusion prematurely -- in a manner that borders on fraud, given the available evidence and lack of certainty at the time. That panel has long since been proved wrong, but one of its co-chairmen (whom we can prove was picked because of his negative bias) and his supporters in the high-energy physics establishment and hot fusion program have been conducting a war against cold fusion. The U.S. Department of Energy spends \$0 on cold fusion, while continuing to dump \$550 million year down the hot fusion black hole. Researchers at federal laboratories have been threatened with dismissal if they dare to work on cold fusion. The science has largely been driven underground in the U.S., in contrast to its open-minded acceptance in Japan.

4. As a result of DOE's actions, no U.S. patents are being issued on cold fusion, while such patents are, indeed, being issued in Japan and Germany.

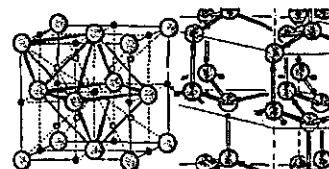
5. One of the main pillars of the anti-cold fusion hysteria in the U.S. was a supposedly "negative" cold fusion experiment conducted by the hot fusion laboratory at MIT in 1989. That experiment has now been proved to be fraudulent and steeped in scientific misconduct. After my resignation as Chief Science Writer for the MIT News Office in June, 1991, because of the unethical behavior I had uncovered, I requested that the misconduct be investigated.

6. Hundreds of scientists working on cold fusion or familiar with the facts about it have petitioned the U.S. House Science, Space, and Technology Committee to hold cold fusion hearings. These scientists, including senior scientists from Japan, have been rebuffed with inaction and neglect, except by a few congresspeople who have taken an interest.

7. Most of the news media in the U.S., with the exception of The Wall Street Journal, Business Week, and an exceptional article by the New York Times in November, 1992 have -- in sheep-like fashion -- bought the negativist DOE line that cold fusion is "dead."

8. If this situation persists and is not exposed, the scandalous difference between the U.S. and Japan situation will make Japan pre-eminent in a field of energy technology that was discovered in the U.S. Furthermore, it is obscene that this revolutionary discovery, which is so pregnant with energy, environmental, and economic salvation for the world, has been derailed in its country of origin.

Eugene F. Mallove, Sc.D. is Author of *Fire from Ice: Searching for the Truth Behind the Cold Fusion Furor*, John Wiley & Sons, 1991



CONTINUED FROM PAGE 3
SUMMARY BY HAGELSTEIN

advance that was pioneered by the SRI group was described: it consisted of the addition of aluminate or silicate to the electrolyte, which caused the formation of a colloidal surface layer that passed light ions (deuterium, lithium, boron,...) and shielded the surface from impurities; this procedure improves the ability of the Pd rod to maintain a high loading ratio.

Two distinct modes of excess power generation were observed; one in which the excess power occurs at relatively low levels (1%-50%) and responds to changes in current density (they have observed 38 occurrences of this mode, lasting hours to many days), and one that is characterized by much higher relative power levels (up to 350% excess) and appears to be insensitive to changes in current (this mode has been observed 3 times, lasting many hours).

For the first mode of heat generation, SRI finds that the excess power rises linearly with current above a threshold current density (which is on the order of 100-200 mA/cm²). A graph illustrating this appears in their conference proceeding. (4) This is in apparent contrast with the Pons and Fleischmann results, which showed a possible quadratic component to the increase above threshold current. Discussion during the meeting pointed to the fact that the SRI experiments are run at constant temperature, while the temperature of the Pons and Fleischmann cells increase when excess heat is produced.

Mckubre presented a graph of excess power production as a function of fractional deuterium loading as determined from resistance ratio measurements. This dependence was found to increase roughly parabolically above a loading of 0.85 ($*P[xs] \sim (x-0.85)^2$) up to loadings near 0.95, which is as high as had been achieved during their C1 experiment.

The group has spent considerable effort chasing down and quantifying uncertainties in the SRI experiments, and are now able to assign meaningful error bars to essentially all quantities measured and inferred in their experiments. The result of this analysis yields rather high sigma numbers on the excess power measurements (in excess of 50 sigma on some of the best data analyzed so far).

Their largest power numbers correspond to on the order of 15 W/cm² (3); it would take a small number of hours of running at this level to defeat a chemical storage explanation. Their highest excess total energy numbers have reached 200 MJ/mole of Pd, which corresponds roughly to 2 KeV per Pd atom; this level of excess energy production cannot be of chemical origin.

3. K. Kunitatsu (5,6) of IMRA JAPAN Co. in Sapporo presented results on their heavy water electrolysis experiments. A number of things struck me as being interesting about this talk, aside from the fact that this is one of the first presentations of anything from this group at a conference at which I have managed to be. This effort appears to have a great deal of resources and some nontrivial technical expertise. They reported excess power measurements as a function of loading (where the loading was determined through measurements of the deuterium gas pressure in the cell), and arrived at essentially the same dependence of excess power on loading fraction as SRI, but with the cut-off shifted down by a few points relative to the SRI results (from a D/Pd ratio of 0.85 inferred from a resistance ratio measurement, down closer to 0.83 determined from measurements of the D₂ gas pressure (6)). The peak excess power occurs when the current density is greater than 100 mA/cm², and the peak excess power which appears on the graph in Ref. 6 is about 35%.

The IMRA experiments differed qualitatively from the SRI and Pons-Fleischmann experiments in that they were run in fuel-cell mode. Conventional Pons-Fleischmann electrolysis experiments are run such that D₂ gas is generated at the cathode and O₂ gas is generated at the anode. IMRA has developed a pressurized cell in which deuterium reactions occur at the anode. Cells operating in fuel-cell mode have been developed in the past, however, this is the first time that I am aware that a Pons-Fleischmann cell has generated excess power sufficiently reliably while operating in such a mode to produce relatively high quality excess power data of the sort presented.

This group seems to have good people, good funding, and much expertise. They presented several poster papers on studies of excess power generation, and the absorption of hydrogen and deuterium in palladium cathodes (7-10) I think that we will be hearing much more from them in the future.

I note that a positive correlation between loading and excess heat production in a Pd/D modified Pons-Fleischmann experiment was reported by Scaramuzzi and De Ninno in a poster paper. (11,12)

4. Ya. R. Kucherov (13-15) from the Luch Association, Podolsk, Moscow Region described experiments that I thought

COLD FUSION TIMES

CONTINUED FROM PAGE 3 :
ROTHWELL

Unfortunately for them, the world is ever-changing, and history will not come to an end until homo sapiens becomes extinct.

6. It is very easy to distort history and to deny the truth by picking and choosing among irrelevant facts, from marginal experiments, and by ignoring the successful work. It is easy to make fun of new ideas, and to attack the integrity of pioneers. Revolutionary new discoveries are always confusing and chaotic at first. Pioneers must make many mistakes before they get anywhere, so they look foolish.

Skeptics put together a mish-mash of accurate but irrelevant facts, the total effect of which create a fun-house mirror distortion. If they were living back in 1907, they might have put together this accurate but incomplete description of events:

The plane was barely controlled on the first flight, that the landing skid cracked when it came down. It was destroyed that afternoon by the wind. At public demonstration a year later in front of newspaper men, the

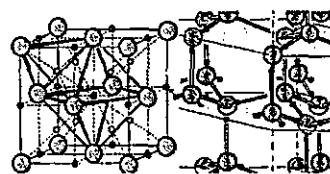
Wrights could not even get the engine to start. In the four years since the "alleged" flight, dozens of skilled workers in Europe and the U.S. have attempted to replicate the airplane, using detailed descriptions from the Wrights, but every single replication failed disastrously. Most experts in the field assert that it is impossible. The New York Times says that a million mathematicians, working for a million years could not solve the problem...

The skeptics of 1907 deliberately chose to ignore massive evidence, and to discount *every* *single* flight after 1903, just as they now ignore all important experiments after 1989. It is very easy to pick just a handful of facts, and to ignore the main point, but it is intellectually dishonest. They deliberately ignore the overwhelming calorimetric evidence of excess heat, and talk only about inconclusive and marginal experiments, and the fleeting evidence of neutrons, which -- for some unknown reason -- only appear in marginal "incommensurate" amounts. In their books, they ignored *all* events in Japan except the Kamiokande work, which was marginal at best. They ignored McKubre.

I believe that all of the "skeptics" understand perfectly well what "megajoules of heat per mole," means, and what it implies. However, they refuse to face the issue. The "skeptics" dance around the issue, and they introduce ridicule, obfuscation, and ignorance into what ought to be an enlightened discussion of one of the most interesting scientific discoveries of this century. They create a smokescreen of nitpicking criticism of the earliest 1989 work, they attack P&F's integrity, and never discuss McKubre, Bockris, or anyone else. They pervert and corrupt the progress of science. In Huxley's words, a skeptic is:

"a man highly endowed by nature and possessed of great means of influence & yet who employs those facilities & that influence for the mere purpose of introducing ridicule into grave scientific discussion."

Jed Rothwell
[Cold Fusion Research Advocates]
(404) 451-9890



CONTINUED FROM PAGE 4 :
HAGELSTEIN SUMMARY

were very important.

The experiment involves using a glow discharge to load a Pd (or other metal) foil (1 cm X 1 cm X 0.1 mm - 1.0 mm) in D₂ gas at 10 torr, with a 400 V discharge (10 - 500 mA current). Apparently this group has had considerable experience with glow discharges and is aware of several tricks that help to preserve the surface of the cathode which helps to attain very high loading (a D/Pd ratio of more than 1).

Numerous effects are observed; excess heat production will first be considered. Temperatures were monitored using W-Re thermocouples in the cathode and anode, and also CC thermocouples in a heat collector some distance from the cathode. Calibration was done through comparing temperature histories of "live" Pd cathodes (cathodes producing neutron, gamma and fast particle emission) in deuterium with those of "worn out" cathodes (cathodes producing no anomalous emission). Excess power production at the level of tens of watts is observed; their best result out of 78 experiments is a 33 watt excess representing a power gain of a factor of 5. Given the small total cathode volume, the resulting power generation rate is quite high; the highest values are on the order of 3000 watts/cm³ of Pd. The highest total energy production observed to date exceeds 20 kJ.

After about 100 seconds after the start of the discharge, neutron emission is observed (a huge signal, reaching up to 10⁶ neutrons/sec in some experiments). The neutron detection described in their earlier work was done using RUP-1 silver activated ZnS scintillation detectors and type SNM-18 gas discharge (13He) detectors. The 10⁶ neutron/sec signal appeared in the scintillation detector as 2000 counts/sec at a distance of 1 meter; the signal showed up as 10000 pulses/second at a distance of 30 cm on the SNM-18 detector. No emission was observed using a hydrogen discharge.

After a while, gamma emission is then observed (also a huge signal, up to 10⁵ gammas/sec in some experiments). The gamma emission was studied using four detectors (Ge-Li, stilbene, NaI and SPS plastic); most of the recent results were obtained using a liquid nitrogen-cooled Ge-Li DGDK-50 detector with 1.6 keV resolution at 1332 keV, and an efficiency of 10⁻³ at 511 keV. An example of an anomalous gamma spectrum from Pd is shown in a recent publication[16].

Gamma lines were identified from short-lived isotopes (the gamma spectrum returns to its initial state in 3-5 days), and some of the identified lines originate in isotopes in the neighborhood of Pd (lines originating from isotopes with a nuclear charge of *Z*-3 to *Z*+8, where *Z*=46 for Pd, were observed).

A very substantial flux (10⁴ to 10⁶ ions/sec) of fast ions is emitted from the cathode, and silicon surface barrier detectors were used for detection. The bulk of the emission occurs between 1-5 MeV, and in some experiments lasts for a few minutes after the discharge is switched off which allows for an accurate determination of the spectrum. Correlated fast ion emission was registered on calibrated CR-39 plates installed inside the discharge chamber.

A small fraction of the fast ions are observed at high energy; peaks were observed at 6 MeV, 12 MeV and 16 MeV. The mass of the particles at 12 MeV and higher was determined to be greater than or equal to 4, as determined through measurements with different barrier thicknesses.

5. There was a Chinese team[17-20] that presented results from a somewhat similar system to that described by Kucherov. A glow discharge was created by applying high voltage (7-11 KV, 50 Hz) between two electrodes inside of a glass bulb containing deuterium at low pressure (4-13 torr). A thin (1 micron) metallic layer of the electrode material (for example, Pd) was deposited on the interior of the glass bulb. The glow discharge current was less than 100 mA; an anomalous current was observed with an average value of 1 A, and excursions up to 10 A. A D/Pd ratio of 0.5-0.8 was claimed to have been obtained.

Substantial neutron emission (13-330 neutrons/sec) was observed, and the energy spectrum was resolved with a recoil proton fast neutron scintillation spectrometer. The resulting neutron spectrum contained both 2.0-2.5 MeV neutrons, and broad emission between 2.5-7.0 MeV; most of the emission occurred above 2.5 MeV.

Neutron emission was also recorded from metals chosen at random, and the signal strength varied with metal according to the order Pt, Nb, W, Pd, Ag, Cu, Mo and Fe. The fluence observed from the D/Pt system was 1.2 X 10⁴ neutrons/sec.

The energy spectrum of the neutron emission for these metals was also observed. In the case of the D/Pt emission shows broad emission up to about 8 MeV, decreasing generally with increasing neutron energy, and with a number of

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CONTINUED FROM PAGE 5 HAGELSTEIN SUMMARY

possible peaks appearing.

Intense gamma spectra were also observed with a NaI scintillation counter during the experiments; the gamma ray yield was about ten times that of the neutron yield. The gamma spectrum of D/Nb showed lines at tens of KeV, 3.4 MeV and 5.8 MeV, and some unresolved emission below 7 MeV.

These experiments seem to me to be similar to the experiment described in 1989 by Wada.[21] Another experiment of this sort was reported by Tazima, Isii and Ikegami, and also by Jin, Zhang, Yao and Wu, at the Como conference.

6. E. Yamaguchi of NTT presented a paper[22,23] on $\{4\}\text{He}$ production from a PdD foil that is sandwiched by gold and $\text{MnO}[x]$. I think of the NTT research labs as being the ATT Bell Labs of Japan, which has an excellent technical reputation. This paper attracted considerable interest in the Japanese media, and there were reports that the price of the NTT stock climbed as a result. The NTT stock climbed a bit more than 10%; Morrison pointed out in his review that the stock went back down to its pre-announcement value within a few days.

In the experiments that he reported, a current of 0.5-0.8 A/cm² is applied perpendicularly to the sandwich. The foil produces heat at a level of 0.5-5 Watts for about 1000 seconds (this is the case whether the foil is PdD or PdH), and then explosively outgasses. At the peak of the outgassing, the samples undergo substantial plastic deformation which lasts for about 10 seconds. During his presentation, it was not obvious whether the temperature rise observed was being claimed as anomalous or not. If the foil is deuterated, these phenomena are accompanied by $\{4\}\text{He}$ emission.

Yamaguchi previously reported at the BYU conference very high levels of neutron emission from this system at a $10\{6\}$ neutrons/second. The experiments described at Nagoya included only helium, heat, and fast charged particle detection.

The $\{4\}\text{He}$ emission is monitored using an expensive high resolution mass spectrometer that is capable of distinguishing between $\{4\}\text{He}$ and $\text{D}\{2\}$ signals, as was demonstrated. A minor peak in the data appears near the expected HT mass position, and Yamaguchi claimed that this signal indicated the presence of HT (Clayton notes in his trip report that the HT signal, if real, would imply a "radiological hazard (> 10 Ci)."). The $\text{H}\{2\}\text{D}$ trimer is more massive than $\text{D}\{2\}$ and does not interfere with the $\{4\}\text{He}$ measurement.

Yamaguchi sees $\{4\}\text{He}$ in his mass spectrometer when he uses PdD, and he sees no $\{4\}\text{He}$ when he uses PdH. Yamaguchi stated that the amount of $\{4\}\text{He}$ was "consistent with the heat," but if he gave figures for the amount of $\{4\}\text{He}$ produced, I missed them. Given that heat occurs for PdH runs as well as for PdD runs, it is not clear what the statement means. Hopefully this issue will be clarified at a later date.

When asked whether the $\{4\}\text{He}$ is due to contamination, Yamaguchi argued that it is not in the $\text{D}\{2\}$ gas used, it is not in the metal, and the vacuum system being used is a high quality system of the type used in semiconductor research that will hold a $10\{-6\}$ torr vacuum for a month without pumping.

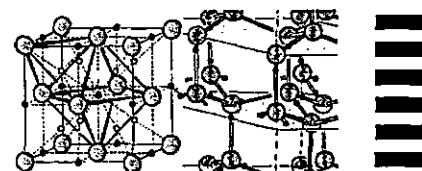
Yamaguchi also sees 3 MeV protons and fast alphas at 4.5-6 MeV using two identical systems based on silicon detectors (Canberra Si-SSD:PD-450-19-700-AM; active area=4.5 cm², active thickness = 700 μ). Protons were observed at 3 MeV, and were attributed to the $\text{p}+\text{t}$ branch of the $\text{d}+\text{d}$ -fusion process. Significant emission was observed between 4.5-6.0 MeV; by comparing signals with and without an intervening 7 μ foil, these signals were identified as being due to either alphas or $\{3\}\text{He}$ nuclei. The total number of fast particles detected was a few hundred per experiment.

The experiment which Yamaguchi and Nishioka have constructed looks very impressive; I got the impression that the helium measurement capability was relatively new. I think that the $\{4\}\text{He}$ signals are real, but I am less convinced yet that it has been made through an anomalous effect. The strongest argument in support of it being genuine is the rather strong time-correlation of the $\{4\}\text{He}$ signal with the temperature excursion of the foil.

The NTT group has been active for years, and by now I think that the basic anomalies which they observe are likely to be right. The new result presented at Nagoya is the helium measurement, which I will be more comfortable with after Yamaguchi and Nishioka have had more experience exploring. I look forward to more results from this group.

I note that the first significant claim for substantial $\{4\}\text{He}$ production in Pons-Fleischmann electrolysis experiments were made at Como by Miles and coworkers at China Lake. Previous negative results had been obtained in searches for helium in the cathode; Miles and coworkers claimed the observation of $\{4\}\text{He}$ in the gas stream. Miles presented a paper at Nagoya[24] which gave an update of the group's recent efforts, which have been hindered by an inability to obtain significant excess heat.

Bockris reported at Nagoya[25] observations of $\{4\}\text{He}$ above background (by factors of 2-100) that accompanied tritium production (described below); the helium was analyzed by thermal expulsion and mass spectroscopy.



7. S. Isagawa et al from the Japanese National Laboratory for High Energy Physics (the KEK collaboration) reported their results on experiments involving searches for heat, tritium and neutrons in Pons-Fleischmann cells.[26,27] I was impressed that the KEK was working at all in this area, and even though they have apparently had an effort at some level since 1989, it appeared to me from their presentation that they have more or less just gotten started.

Most of their results to date are negative, and it appears that they are confident that they are going to get the expected (that is, null) results. Of the possible excess heat events that they have observed so far, they have been able to rule out all but one as being due to known (non-anomalous) causes. Neutron emission is mostly not observed, but they have one event at 3.5 sigma of excess neutron emission (23 ± 7 neutrons/sec) recorded over nine hours from one PdD cell after 20 hours of electrolysis.

Although their results to date have little impact on the field, should they continue, their contributions could be and should be substantial in time. One thing that this group would be able to do which few other groups in the field are as well suited to do to bring on board the physics community. Positive results obtained at the KEK would stir interest in other physics laboratories as almost no other result. The physicists have written off Pons and Fleischmann, so they are free to ignore the claim of kilowatt/cm³ reported at the conference; but if the KEK gets 10% heat power at 10 Watts/cm³, I would bet that every physics lab on the planet will likely be pulling out their electrochemistry sets again.

8. T. P. Perng of National Tsing Hua University in Hsinchu, Taiwan described observations of excess power from molten salt electrolysis experiments as part of a paper on heavy water Pons-Fleischmann experiments.[28] Liaw and coworkers at the U. of Hawaii described at the last two international cold fusion conferences experiments using molten salt electrolysis with Pd and Ti anodes in a LiCl-KCl eutectic saturated with LiD. At Como, the group reported the observation of excess power at a level of about 10 times the input electrochemical power in Pd (up to a 30% increase over electrochemical plus heater power), with an energy gain reported as about 1 GJ/mole Pd or 6 MJ/mole D₂. The Pd anode volume was 0.040 cm³, so that the excess power per unit anode volume is about 250 watts/cm³.

This result was important because the power excess was so large relative to the electrochemical input power, and because the temperature excess was on the order of a hundred degrees centigrade, which would have the potential for efficient energy extraction.

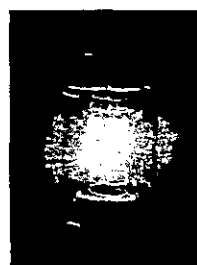
Little progress has so far been reported toward a reproduction of the Liaw experiment. Perng described results from an experiment performed to provide a confirmation of the Liaw experiment; the power excess claimed was on the order of 2-5 times the input power.

The preprint[29] from C. M. Wan et al consists of an abstract and copies of 11 figures, from which I will attempt to give an account of the work. Following Liaw, the palladium electrode is used as an anode (instead of a cathode as is done in conventional Pons-Fleischmann experiments), immersed in a KCl-LiCl eutectic saturated with LiD. The anode dimensions are 6 mm diameter and 5 cm length. The molten salt sits in an aluminum container which serves as the cathode. The temperature is sensed using a thermocouple embedded in a quartz tube which is placed in the molten salt in the general vicinity of the Pd anode. A Ni-Cr alloy resistive wire heater (encased in quartz) is wrapped around the cell, within a ceramic fiber insulator. Nearby is a $\{3\}\text{He}$ neutron detector with a 0.01% detection efficiency.

From the figures and the abstract, it is clear that a time-dependent excess neutron signal appears following 200 hours of electrolysis, at twice background (background is 5.51 ± 0.44 cpm) corresponding to 800 neutrons/sec. This neutron signal is rather clearly correlated with the excess power production which is time-dependent and rises to about 10 watts. Given the large anode volume, this level of excess power corresponds to about 7 watts/cm³. The associated temperature excursions are about 25 Deg. C, with one excursion up to 50 Deg. C.

The abstract quotes power gains of 5 to 108 for the 6 mm Pd rod which is 5 cm long, and power gains of 8 to 560 from a 4.5 mm diameter rod. These numbers are very high and represent excesses in comparison with electrochemical power

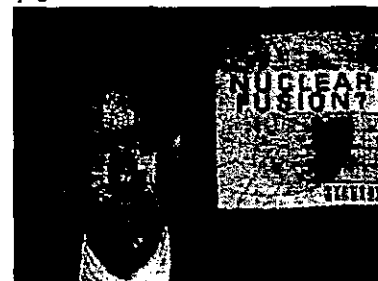
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FOUR YEARS AGO

Left: Cover of Fire from Ice (E. Mallove)

Right: Dan Rather, CBS News announcing "Cold Fusion" in "the March" of 1989



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rather than total input power; I think that the highest numbers correspond to modest excess powers observed at low input current level.

There was an abstract from the National Tsing Hua University by Yuan et al that described a molten salt experiment; [30] I do not know whether this paper was presented. I suspect that this paper may not have been presented, and that Perng was reviewing results obtained by his colleagues.

9. There were several papers on attempts at replicating the Takahashi experiment that captured the attention of the Japanese press earlier this year. A. Takahashi described earlier this year obtaining tremendous excess heat in a heavy water electrolysis experiment that ran at an average of 1.7 output power over input power for about two months. [31,32]

The total excess enthalpy generated was claimed to be about 2250 MJ/mole Pd (more than 20 KeV per Pd atom), which is one of the highest claims to date from this type of experiment. Following Takahashi's announcement, many laboratories attempted a replication.

Takahashi's experiment is similar in many ways to the classical Pons-Fleischmann experiments with some variations. A Pd foil from Tanaka Kikiniku Kyogo (Tanaka Precious Metals Co.) with dimensions 2.5 cm X 2.5 cm X 1 mm is used for a cathode instead of a rod. The electrolyte volume is very large (700 cc of D₂O with 0.3 M LiOD). An innovation of Takahashi is the use of a time-varying current which alternates between a high mode (4-5 amps) and a low mode (0.2-0.4 amps) every six hours.

Takahashi's calorimeter is an open cell flow calorimeter, where water from a chiller is flowed through a coil inside the cell, and the power is determined from a knowledge of the mass flow rate and the input-output temperature difference. An advantage of this type of calorimeter design is that it is able to function at near constant temperature when high power is applied to the electrochemistry. The temperature was monitored using teflon coated thermocouples at the inlet, outlet, and cell interior. The cell was calibrated before and after the run in the initial experiment, and the calibration lines were approximately reproduced.

The total input energy for the initial experiment was 250 MJ, the total output energy measured was 410 MJ, leading to an excess of 160 MJ. Takahashi's excess power level claimed was 32 watts averaged over two months, with excursions to 100-130 watts. The cathode volume is 0.625 cm³ (0.0706 moles), so that the average power density is 51 watts/cm³ and peak excursions are 160-208 watts/cm³. As discussed above, a chemical explanation of the effect (barring other systematic errors) is defeated in less than half an hour at the high excess power levels.

Attempts at replication had varying degrees of success, but no one has been able to reproduce the very high power levels claimed by Takahashi. In the reproduction which Takahashi reported at Nagoya, the excess average power was 8 watts, with excursions to 15 watts.

Takahashi's method comes with the recommendation that it is a potentially technically easier experiment than other experiments which have been reported. It is less exacting in the rigorous electrochemical purities require compared to the SRI experiments; the cathodes from Tanaka metals are readily available to workers in the field, in contrast to the Johnson-Matthey cathodes employed by Pons and Fleischmann. The system is in principle relatively cheap to set up; Mallove [33] described a version of the Takahashi experiment which was built up using about \$10K of hardware funds.

Storms at LANL is claiming excess heat from a Tanaka batch 1 foil, and no heat from a batch 2 foil. [34] The anomalous power from the first foil was more than 20%. The batch 2 foil suffered an increase in internal volume on loading, which Storms suggested might be used as an indicator of whether a cathode was suitable for heat experiments. No one has reported a success with a batch 2 foil that I am aware -- batch 3 appears to be free of this problem.

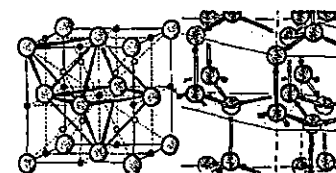
Celani [35,36] described efforts to reproduce the Takahashi experiment at his laboratory in Frascati. Experiments were run in an open cell flow calorimeter using two Tanaka metals batch 1 cathodes, one Tanaka batch 2 cathode, and one IMRA batch 1 cathode. Positive results were obtained with the Tanaka batch 1 rods (at 8% and 25% peak power excess) and with the IMRA rod (12% peak excess). No excess power was observed with the Tanaka batch 2 cathode. Blank experiments were performed where a gold cathode sheet was substituted for the Pd cathodes, and no excess was observed. A correlation between high loading and excess heat was noted; small amounts of excess tritium were reported for the runs which gave excess heat.

Oyama [37] reported a 2.4% excess energy, which is small, but was measured with much smaller error bars; a light water blank showed no excess.

10. Tritium production was discussed by several groups. The existence of such an effect is interesting because it constitutes an additional signature of the presence of a nuclear phenomenon; tritium cannot be made chemically. An additional feature of many tritium experiments is that the tritium is not accompanied by neutron emission (neutron/tritium ratios of 10⁻⁷ - 10⁻⁹ have been reported). The *dd*-fusion reactions would produce neutrons and tritons in roughly equal amounts, so that the observations imply either a new mechanism or else a very significant modification of the fusion reactions. Possibly more significant is that 14 MeV neutrons from *d-t* fusion reactions would be expected from secondary reactions if the tritium nuclei were created with MeV-level kinetic energy. The very low neutron to tritium ratios claimed imply a very low triton energy (below 10-15 keV), sufficiently low to be inconsistent with all but the most exotic reaction mechanisms.

At Como, strong presentations of tritium production were made by Will, Claytor, Lanza, Szpak. Will has not been active in the field during the past year, and did not attend the Nagoya conference; Lanza has continued, but was not able to attend this year; Szpak has continued, but was also not present at Nagoya.

Claytor [38] (whose work impresses me) described further experiments on tritium production in which a stack of alternating layers of palladium and silicon is placed in deuterium gas at over 10 atmospheres, and a pulsed current is passed through the stack. Reproducible tritium production is claimed at levels of 0.02-0.2 nCi/hr (1.1 X 10⁶ - 1.1 X 10⁷ tritium atoms/second). Advances which the LANL group has made during the past year includes: reduction of background tritium, improvement in detection sensitivity, improvement in



reproducibility at higher tritium generation rates, and the innovation of working with stacks using Pd built up from powder. Upper limits on neutron emission can be placed from their work as reported at Como of 4 X 10⁻⁹ neutrons/second; the (3)He neutron detector which they have used (which I saw during a recent visit there) has a roughly similar efficiency for 14 MeV neutrons as for 2.45 MeV neutrons (the detector is more sensitive at 2.45 by a factor of 1.5 according to Menlove).

Bockris [25] described two tritium experiments at Nagoya. In one experiment, a reproduction of the Szpak-Boss experiment described at Como was attempted. Szpak codeposited Pd on either a Cu or Ni substrate from PdCl₂ in D₂O containing 0.3 N LiCl; continued electrolysis resulted in tritium production. The Pd was observed to plate out during the first 6-8 hours, and excess tritium would be detected about 10 hours later. The experiment Bockris reported involved codeposition of Pd on gold, and tritium production was observed to start as soon as 10 hours after the Pd deposition, and production up to 3 times background was observed. This is the first successful confirmation of the Szpak experiment of which I am aware. Bockris did not obtain the high degree of reproducibility claimed by Szpak.

I note that Miles also described [24] attempts to reproduce the Szpak experiment, and reported the observation of a modest tritium increase, but "not clearly beyond levels expected for electrolytic enrichment due to isotopic separation factors."

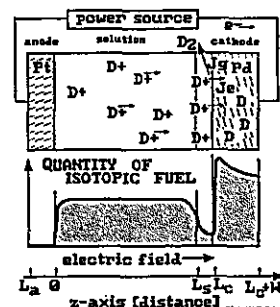
Much more spectacular are the results reported by Bockris of Chien's experiments on tritium production in a Pd electrolysis experiment where massive amounts of tritium (more than 10¹⁵ atoms) were observed. This report is very significant because it represents a new claim for very high levels of excess tritium.

Early on, reports of very high levels of tritium were reported to have been observed at Texas A&M, corresponding to production rates on the order of 10¹⁰ tritium atoms/cm³ or higher. These experiments were clouded by charges of fraud (this charge was apparently investigated by a panel hired by Texas A&M, and not substantiated), and by the observation of high levels of tritium contamination in Pd claimed by

continued on page 8

"In March 1989 electrochemically induced nuclear fusion reactions were reported, but were initially very difficult to reproduce. By 1992 several experiments have demonstrated that anomalous nuclear processes occur in palladium which has been highly loaded with deuterium.

The application of an electric power source generates an applied electric field intensity. Within the solution the deuterons are distributed mainly tightly bound to oxygen atoms as heavy water. Full loading appears to be a prerequisite, but possibly insufficient, requirement before fusion occurs."

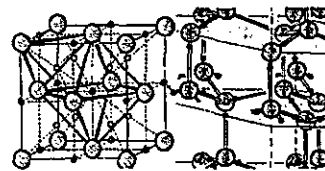


after M. Swartz, Fusion Technology, vol. 22, pp 296-300 (9/92)

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Wolf (found by dissolving Pd rods in acid, and then performing scintillation counting on a neutralized version of the resulting solution). Similar experiments reported by Cedzynska (differing in that the distillate of the solution was analyzed by scintillation counting) failed to show contamination at the high levels reported by Wolf, and in addition found that false positives could occur when the solution was not first distilled.

Subsequently, much more stringent controls were done to attempt to defeat the insidious tritium contamination claimed by Wolf, and new post-Wolf experiments were reported in which anomalous tritium production has been claimed. Except for experiments reported by BARC, most new claims have involved tritium production rates many orders of magnitude below those of the initial claims (the new claims generally ranged from 10^4 - 10^6 tritium atoms/second). The significance of the Chien experiments is that the tritium production rate claimed by Chien (in a post-Wolf experiment) are some of the first to approach the very high early (pre-Wolf) experimental claims. Much care was taken to avoid possible contamination: samples from the same rod were dissolved in aqua-regia, and the resulting solution distilled, and then analyzed using a scintillation cocktail following the method described by Cedzynska at Como.

Chien's earlier experiment was carried out at the Institute of Nuclear Energy Research in Lung-Tan, Taiwan. The palladium cathodes used were 1.0 cm in diameter, and 1-2 cm long; Pt wire wrapped around at a distance of 4 mm was used for the anode. Electrolysis was carried out in heavy water with 0.1 M LiOD; tritium assay was done with a scintillation cocktail. Solutions exhibiting high tritium activity were sampled at the time of the experiment (10/89), and then resampled 10 months later (8/90) in order to observe the tritium decay from the sample. Tritium generation rates of 10^6 - 10^9 atoms/second were determined, lasting for a total of 20-30 days. The numbers claimed by Bockris at Nagoya[25] for the Texas A&M version of the experiment correspond to about 10^7 atoms/sec/cm² of surface area, in experiments with a 3-6 cm² surface area.

11. V. A. Romodanov gave an oral presentation in the theory panel at Nagoya.[39] Romodanov's command of the English language was imperfect; he read from his paper for more than 20 minutes in a thick Russian accent. Essentially no one with whom I talked understood the point of what he said, and his abstract did not particularly add to the information content. Given that his talk occurred in the theory section, and given that his theory appeared to be largely classical fusion modified somewhat by lattice effects, no one was expecting that a major experimental result was buried in his presentation. Two things about his talk raised flags for me, indicating that I should try to follow up if possible. One was that he was from Luch, which is the same place Kucherov is from...and I was very impressed by Kucherov's results. The second thing that I recall was that there was a table giving some very high tritium numbers; at the time I thought they were theoretical estimates because they were so large.

Romodanov handed me a preprint[40] which explained in rather clearer English what was the content of his talk. I will focus on what I consider to be the single most important part of his presentation, which if true, is of fundamental importance. Romodanov described the results of glow discharge experiments which appear to have been done on a system very similar to that discussed by Kucherov (see above in this review). Romodanov and his colleagues focused on the detection of tritium produced in glow discharge experiments in Pd and in other metals.

The glow discharge was run in deuterium gas at 100-200 torr, with an applied voltage in the range of 40-125 V, and a current of 3-4 A (a wide range of operating conditions are described in the paper, and the numbers I have chosen appear on one table -- I am not completely certain from the paper that the tritium generation was done with these parameters). Various cathode metals were used, including Y, Mo, Nb, Er, Ta, and W; as disks with a diameter of 13 cm and a thickness between 500 μ m and 1 mm, or rods of 0.5-2 cm diameter. The cathode temperatures were measured to be between 970 Deg. K and 1670 Deg. K, with only minor (15% or less due to anomalous self-heating effects).

Tritium generation rates between 10^5 atoms/second and 10^9 atoms/second were measured in the different metals under various conditions. The largest rate (1.7×10^9) was obtained in Nb at 1170 Deg. K, corresponding to an increase in tritium activity in the deuterium gas of 2.3×10^4 . The neutron emission was measured in these experiments with a "radiac instrument RUP-1," which appears to be a scintillator with silver activated ZnS dispersed in transparent plastic (sounds similar to the detector used by Kucherov), and a neutron to tritium ratio of 1.8×10^{-7} was obtained.

12. R. Notoya from Hokkaido University brought a light water demo that was set up and operated in the hallway of the conference[41]. The demo consisted of two cells: in one cell was a resistive heater, and in the other cell was a nickel cathode immersed in a light water K₂CO₃ electrolyte, similar to the method of R. Mills and colleagues.[42] Notoya's method differs from Mill's method in that (1) the Notoya cathode is made of porous nickel, and the Mills cathode is plain nickel; and (2) Mills uses an intermittent current, while Notoya uses a constant current.

The resistive heater was driven at 2.1 Watts electrical *iv* input; the electrolysis

cell was driven so that the joule heating in the cell was also 2.1 Watts. The *iv* input into the electrolysis cell is actually higher by about 30%, but since electrolysis is occurring with a Faradaic efficiency near unity, the power ending up inside the cell is matched as long as no recombination occurs in the space above the electrolyte. The live cell ran higher by about 15 Deg. centigrade than the blank, as could be inspected visually by observing alcohol thermometers immersed in both cells. Notoya claims that the light water cell temperature implies a factor of about 3 more net power input, or roughly 6 watts of heating present.

This was interesting for a number of reasons. This was the first live demonstration of excess heat production at a cold fusion conference that I am aware of. I have always thought that live demos would start to show up at conferences and at presentations, but I had figured that the first ones would be heavy water demos. I thought that it was significant that Notoya's system works well enough for her to be willing to bring it as a demo at a major international conference.

Many of the "established" workers in the field who have put in substantial effort on heavy water Pons-Fleischmann cells and have observed heat simply do not believe that a heat effect can be observed in light water. Among other arguments that can be heard is that if the effect is either nuclear or is fusion, it must involve deuterium. Others in the field argue that the light water claims are simply due to sloppy experimental work. Independent of the correctness of the various assertions, it is almost humorous to find senior members of the coldion community sounding very much like their critics and tormentors of 1989.

The first reports of heat from light water experiments were actually from Pons and Fleischmann early on in 1989; when I last spoke with Fleischmann about light water experiments about a year ago, he was firm in his conviction that it was not possible to get excess heat from a light water cell.

R. Mills, who is the originator of this particular Ni/K₂CO₃ experiment, has no previous reputation or standing as an electrochemist, calorimetrist or physicist. He rejects the notion of cold fusion as due to nuclear effects completely (in fact, he does not wish to associate himself with the cold fusion community, and does not consider his effect to be related in any way to cold fusion), and has developed his own theory as to why his experiment works; his theory is based on the proposed existence of orbitals of hydrogen that lie below the 1*s* level. An explicit assumption in the Mills and Farrell theory is that the electronic charge distribution in hydrogenic states consist of charged shells of infinitesimal thickness. In order for this proposal to be correct, quantum mechanics must be incorrect (which Mills believes -- he offers his theory as a replacement for quantum mechanics). There have been no observations of such states, and the existence of such states would likely not be consistent with the observed stability of atoms as atoms.

Reproductions of the Mills experiment have been reported previously. Noninski[43] published positive results from his experiments; Noninski views his experiment as a verification of the work of Fleischmann and Pons, who state explicitly (in a 1989 patent application) the possibility of excess heat in a light water cell with Ni as a cathode material. Mills was apparently unaware of Fleischmann and Pons patent application and its relevance.

Confirmations of the Mills light water experiment have also been reported by Srinivasan,[44,45] and Bush and Engleton.[46] Notoya and her laboratory come with good reputations; her confirmation of the Mills experiment (complete with demo) is probably the most significant endorsement of the light water excess heat results standing.

Notoya's demo is an open cell system. It operates at a sufficiently high excess power that recombination or other effects that would make an open cell system perform differently from a closed cell would not change the essential result even if the recombination and other secondary effects were taken into account incorrectly or ignored. You can put your finger on the tubes Notoya's demo to convince yourself that a very significant temperature difference occurs. This was also claimed for the Mills experiment, as well as for other experiments reproducing the Mills result.

During the conference and afterwards, a virtual firestorm of controversy arose concerning the difference in wires that were attached to the live cell and to the blank. A student of Steve Jones suggested that since smaller diameter wires were used on the blank, that the reduced voltage drop across the resistor could account for the difference. After the conference, Notoya replaced the offending wires, and reported essentially no difference in the resulting blank

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"... after a few more flashes
in the pan, we shall hear very
little more of Edison or his
electric lamp. Every claim he
makes has been tested and
proved impracticable."
[New York Times
January 16, 1880]

"Professor Goddard ... does
not know the relation of
action to reaction ... he only
seems to lack the knowledge
ladled out daily in our high
schools"
[New York Times
January 13, 1920]

COLD FUSION TIMES

CONTINUED FROM PAGE 8

PETER HAGELSTEIN'S VIEW OF THE 1992
NAGOYA CONFERENCE CONTINUES

temperature.

Notoya's demo was brought to the US and set up at MIT during the first week of December. During the day and a half before her presentation at MIT, the live cell and resistive blank ran at very nearly identical temperatures, consistent with no excess power production. Notoya attributed this to contamination of the nickel cathodes. After her visit, she returned to Japan and set up her demo in a laboratory in Tokyo where excess heat was observed. A few days later, she was back at MIT, attempting for a second time to demonstrate excess power production. During this visit, a temperature excess was seen during electrolysis of the second cathode tried. According to Notoya, the initial temperature differential in this case corresponded to a 100% power excess. Subsequently, a persistent excess of about 4 Deg. C was observed, which she said corresponded to a 30% power excess (the reduction in fractional excess was attributed by Notoya to contamination).

The persistent excess power which Notoya obtained at MIT was about 0.75 Watts, and the cathode volume was about 0.05 cm³, leading to a volume averaged excess of 15 Watts/cm³. At Nagoya, the cell ran at a 4 Watt excess, corresponding to 80 Watts/cm³. She claims that she has observed a maximum of 200 Watts/cm³ excess. A few hours of operation at 15 Watts/cm³ is sufficient to defeat a chemical explanation, which was done at MIT. The power excess demonstrated at Nagoya would defeat a chemical explanation in tens of minutes, and the cell ran for many hours.

Little is known about loading ratios (H/Ni) while heat is produced; no information is available about potassium loading in the Ni; there is apparently an alkaline intermetallic layer formed which is at least several hundred Angstroms thick which may play a role. Nothing is known about the temperature sensitivity of the effect; Notoya observes the excess power to be essentially linear in applied current down to her lowest values (50-100 mA, and about 1 cm² geometric area at MIT; the high current levels approach 1 amp).

Notoya obtains her best results with cathodes which have an extremely high area ratio (real area to geometric area), and she uses cathodes with an area ratio of several thousand (and a reduced density of about 6 gm/cm³). The effect is apparently extremely sensitive to contamination, especially to oils. She observes an increase of 20% in calcium concentration (near 20 ppm) in the electrolyte, which she believes may be anomalous.

I do not think that there is yet any particular contradiction between the light water experiments of the Mills type and the light water blanks in Pons-Fleischmann experiments. The light water blanks in Pd/H experiments run in a H₂O/LiOH electrolyte give zero excess power in most everyone's blank experiments these days; the Mills experiment uses a Ni cathode with a H₂O/K₂CO₃ electrolyte. These are really very different systems. In any event, in time any connections between the two systems will be clarified.

Based on Notoya's work, the evidence in support of a light water effect has improved significantly. The effect which she observes is so great that there appears to be no simple explanation for it.

So is there a light water effect? At this point, I am not yet sure one way or another. On the plus side: (1) the effect is large, (2) looks to be nuclear given the excess heat numbers, and (3) can be reproduced. On the minus side (from my point of view): (1) the effect has been studied by a relatively small number of groups for a relatively short time, (2) the effect appears to be somewhat insensitive to choice of electrolyte (claims of 41) of heat production have been made for experiments which have used other alkali-carbonates such as Li₂CO₃, Na₂CO₃ and Rb₂CO₃ and to some degree the choice of cathode (positive results were reported for Ni, Ag, Au and Sn electrodes). The reason which I am uncomfortable with the insensitivity of the effect to cathode and electrolyte comes from potential difficulties associated with finding a reaction mechanism that would show such an insensitivity.

The experimentalists have grown used to the idea that deuterium gives anomalies and hydrogen does not; the theorists who believe in fusion mechanisms are comfortable with positive effects in deuterium and negative effects in hydrogen. A light water heat effect causes consternation in both camps; it would be exceedingly difficult to reconcile with a fusion mechanism.

The neutron transfer model which I have been looking at (described briefly below) needs a neutron donor (usually deuterium) and an acceptor nucleus, and therefore has somewhat fewer constraints; nevertheless, I do not relish the prospect of attempting to explain an apparently general light water heat effect where the nuc-El present are widely different from one cell to another. An experimental determination (and confirmation) of the ashes in any of these experiments would of course greatly improve the situation.

As a result, I am not yet sure that there is a light water effect. I will be sure one way or another when more confirmations (or non-confirmations) are reported. I will be sure when Notoya, who has worked on her experiments only since last August, has had more time to think about her experiments and to improve them. I will also be sure in time after the cold fusion community has had more time to study and to evaluate the experiment.

13. The successful production of significant excess energy must give rise to ashes of one sort or another. It is not currently known what reactions are occurring; consequently, it is not obvious what ashes are to be expected. Energy excesses in the range of 1 MJ to 10 MJ have been reported in several experiments; we will consider briefly the implications of excess power generation, both per joule and for a 10 MJ total excess.

Conventional *dd*-fusion reactions producing 10 MJ would yield more than 10¹⁹ neutrons, and a roughly equal number of tritium atoms. Pons and Fleischmann's recent measurements (48) of the neutrons produced from their cells yield 5-30 neutrons per joule, low by more than 10 orders of magnitude from what would be predicted for conventional *dd*-fusion. Tritium is not produced in their experiments, with a limit which is probably on the order of 10¹⁴ tritium atoms per joule; low by at least 8 orders of magnitude.

It has been suggested that the [4]He branch of the *dd*-fusion reaction is somehow favored, and several searches for [4]He have been made. The conventional [4]He branch yields a 24 MeV gamma, which is not observed when heat is produced. The reaction energy would have to go elsewhere to be qualitatively consistent, and many in the field believe that energy transfer to the lattice occurs. Many measurements have been performed seeking [4]He in the cathode after the experiment; my impression is that it is simply not there quantitatively by many orders of magnitude.

There have been some efforts seeking [4]He in the gas stream produced during electrolysis; Miles focused the attention of the community on this issue last year at Como when he claimed the observation of [4]He which at its highest levels might account for roughly 10% of the excess energy. Scaramuzzi and De Nino (11,12) described a new cell, calorimeter plus helium detector with which they plan to attempt a confirmatory study. Other groups are acquiring mass detection capability for similar studies. The measurements of Yamaguchi (22,23) described above has also raised interest in helium detection.

I would think that by next year's conference, that there will be a consensus by many groups established on whether substantial helium is produced or not.

If the pragmatic point of view is adopted that whatever reaction is occurring is not constrained by theoretical preconceptions, then the search for the ashes is generalized considerably to include possible isotopic shifts or anomalies, and the possible production of elements or isotopes not initially present.

A large number of studies have been reported at the international conferences in which the cathode surface has been analyzed for the presence of trace elements. Due to the nature of electrochemical deposition in real systems, quite a long list of surface contaminants are found at significant levels, hopelessly complicating any straightforward *ab initio* experimental search.

The number of nuclei which is sought is on the order of 10¹² per joule (or 10¹⁹ per MJ), which either helps or hurts depending on the point of view. Present in large quantities are D, O, Pd, Li, and H. Determining a relative isotope shift between deuterium and hydrogen is generally deemed not to be feasible, given the presence of hydrogen as a ubiquitous universal contaminant. Isotope shifts in oxygen are not currently predicted by anyone in the field, and have never been studied in Pons-Fleischmann experiments to my knowledge.

The first serious claim of possible isotope shifts in heavy elements with which I am familiar was made by Rollison at the NSF/EPRI workshop in 1989 (Rollison subsequently had to back down from her claim - see the proceedings of the Salt Lake City conference). The glow discharge observations described by Kucherov (13-16) and the Chinese group (17-20) imply isotope shifts in Pd and other metals.

The production of 10 MJ of energy in a Pd cathode (containing 0.1 mole of Pd) would give rise to modifications of the Pd isotope distributions (assuming Pd were fuel) at the 0.002 level, assuming arbitrarily 5 MeV per reaction. The prospect of proving this experimentally if it is in the bulk is judged to be impractical. If the reactions occur near the surface, then the numbers improve; the "noise" associated with natural isotopic separation also increases. Searches for such surface isotope shifts have been reported, and continue to be performed; such searches for now remain in the background of the field.

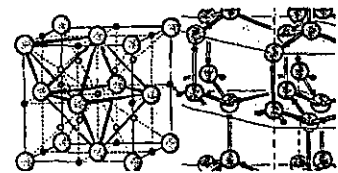
Lithium appears to be required for heat production in Pons-Fleischmann experiments, although it is unknown whether it plays any nuclear role.

Thompson, formerly of Johnson-Matthey, reported (49) that the lithium on the surface of a Pons-Fleischmann cathode that had been involved in heat-producing experiments showed a depletion of [6]Li relative to the natural abundance (down to about 4%). Pons and Fleischmann had reported (in the Salt Lake proceedings) that the lithium which they had used was initially enriched in [6]Li (11%); Thompson noted this in his talk, quoting an initial concentration of 9-11%, but said the Johnson-Matthey group did not have a before to compare with their after.

During the questions following Thompson's talk, McKubre noted that the Johnson-Matthey analysis only looked at the surface, and that any lithium present in the bulk might provide an internal reference. Thompson said that he thought that this suggestion could be tested. The conference proceedings from Johnson-Matthey states that no lithium is detected in the bulk, [50] which would imply that it will not be possible to establish an internal reference retrospectively.

The amount of lithium present in a cathode is an interesting question. Gozzi reported last year the results of studies to determine Li loading in Pd during electrolysis, and found the very high number of 5% by monitoring the Li lost from the electrolyte. I questioned him at the conference (he presented [50] some nice positive results from his torus of cells where he monitors for heat, neutrons, tritium, helium, and I think gamma; unfortunately, I am lacking sufficient documentation of his results to present more details in this review), since several papers presented at each of the international conferences showed per cent level surface concentrations which fall rapidly on the micron scale into the bulk of the cathode (an exception to this was the measurements presented by Nakada [51] et al showing lithium profiles with significant lithium in 20-30 microns). Lithium concentrations were measured by Myamoto et al. [52,53] who obtain Li/Pd ratios between 3 X 10⁻⁴ and 3 X 10⁻³. I suspect that the Li/Pd ratio is probably sensitive to cathode properties, to the electrochemistry and loading time; one possible explanation of the long loading time required for Pons-Fleischmann cells, and remarked on explicitly by McKubre, (2) was that extra time beyond the deuterium loading time required to see the heat effect might be due to a necessity to achieve significant loading of another species, such as lithium or other light interstitials.

I note here that energy production at the level of 100 MJ/mol would yield an observable (2%) isotope shift in lithium if the lithium



9

concentration were at the 1% level, and if the bulk lithium did not substitute with lithium in the electrolyte. The numbers are worse if the electrolyte lithium is included, but not so bad to prohibit a measurement. Unfortunately, very few groups are currently pursuing the lithium isotope shift problem; I consider it to be an important question, especially in light of the initial Johnson-Matthey positive measurement.

14. B. Stella (54) presented a poster that I passed by twice; the title talks about the "stimulated emission of neutrons," that is of course impossible -- neutrons are fermions, and can of course not participate in stimulated emission. The third time by, Stella grabbed me and walked me through his poster (for which I am thankful, otherwise I would have missed it).

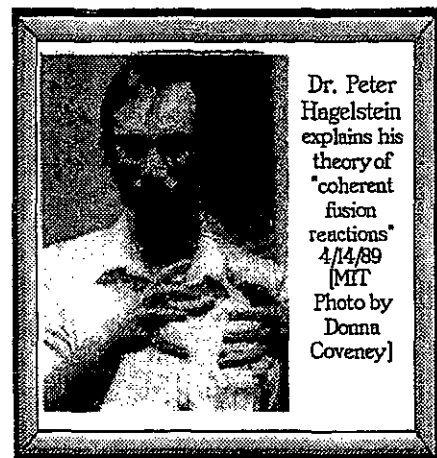
In essence what the experiment consists of is taking a Pons-Fleischmann cell, putting it inside a 40 % efficient neutron detector underground in the Gran Sasso INFN laboratory, and directing an incident neutron beam (of about 30 neutrons/sec) with a substantial thermal component at the cell. Fast (2.45 MeV) neutrons are measured originating from the cell, and a gain of about 2 fast neutrons for every incident neutron is claimed.

After talking to him, I was given to understand that for 30 neutrons/sec input that 60 or more neutrons/sec were measured (taking into account the neutron detector efficiency). I asked if the ratio held up at a higher input flux, and he said that they had done experiments up to 500/sec incident, with the same basic neutron gain (but that their neutron detector suffered from saturation problems at such a high flux).

I asked whether the effect was reproducible. Stella said that they had done two runs (each run takes about a week to do) so far, and that they hoped to be able to do some more in the near future.

I note that a neutron gain of 2 would be a very important result, if true, with rather important implications. I note also that this report of two observations (with a modest signal relative to noise) is the first in which such an effect has been claimed, and that no reproductions from either this group or any other group has been made.

===== end of Part I (Part II to follow) =====



Dr. Peter Hagelstein explains his theory of "coherent fusion reactions" 4/14/89 [MIT Photo by Donna Coveney]

Prof. Peter Hagelstein teaches at the Massachusetts Institute of Technology, Electrical Engineering and Computer Science (VI). He is the inventor of two types of x-ray lasers.

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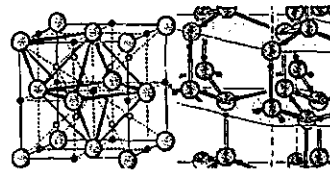
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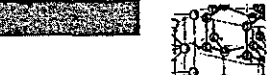


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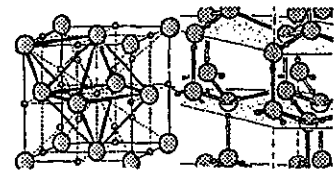
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PUBLIC COLD FUSION PATENTS/APPLICATIONS

This sorted list was compiled based upon data provided by Hal Fox of Fusion Facts (Utah). The February 93 issue of Fusion Facts discusses these in more detail.

DATE	NUMBER	INVENTOR
89/03/13	Iwo9010935	Pons
89/03/31	Ius4986887	Gupta
89/04/04	Ide3910806	Hora
89/04/05	Ijp90276991	Hitachi
89/04/05	Ijp90276989	Hitachi
89/04/06	Iwo9013129	Hagelstein
89/04/06	Ijp90267495	Jap Steel
89/04/10	Ijp90276990	Hitachi
89/04/13	Iep392325	SEL(Japan)
89/04/13	Iwo9013126	Mills
89/04/13	Ijp90271288	Mizugai
89/04/13	Iep392324	SEL(Japan)
89/04/14	Ijp90276992	Hitachi
89/04/18	Iwo9013127	Ceramatec
89/04/18	Iep394204	ENEA(Italy)
89/04/19	Ijp90278189	Tanaka
89/04/20	Iep393463	SEL(Japan)
89/04/20	Ide3913002	Dies
89/04/20	Ijp90280088	Sanyo
89/04/23	Iwo9013128	EPRI
89/04/26	Ijp90285283	Bridgestone
89/04/27	Iep395066	Matsushita Elel
89/04/27	Iep394980	Matsushita
89/04/28	Ijp02287289	Sumitomo
89/05/01	Iwo9014668	Cravens
89/05/02	Iwo9014670	EPRI
89/05/09	Ide3916397	Siemens-AG
89/05/09	Ide3915153	Siemens
89/05/09	Ijp90293692	AIST
89/05/10	Idd295939	Hum. Berlin
89/05/12	Iwo9013897	Drexler Tech
89/05/12	Ijp90302693	Sanyo
89/05/12	Idd293148	Tech Dresden
89/05/12	Idd293147	Tech Dresden
89/05/18	Ijp90304393	Seiko-Epson
89/05/19	Ijp90306194	Sanyo
89/05/22	Ijp90307093	Seiko-Epson
89/05/29	Ijp90311792	Seiko-Epson
89/05/29	Ijp90311793	Seiko-Epson
89/05/31	Ijp91002690	Hirokasu
89/06/02	Iwo9015415	Johnson-Matthei
89/06/05	Ijp91006490	Nippon Light MI
89/06/05	Ijp9107113	Sanyo Electric
89/06/05	Ibe1002781	Bogaert
89/06/06	Ifr2647943	Forrat
89/06/14	Iep402988	Ecoline
89/06/14	Iwo9016070	Cond. Matter T
89/06/21	Ide920312	Siemens-AG
89/06/23	Ijp91025393	Hitachi
89/06/23	Ibp91025392	Hitachi
89/06/30	Iwo9102360	Schoessow
89/06/30	Ijp91033687	Taiyo Yuden
89/07/03	Ijp91035193	Nuc Fuel
89/07/03	Ijp91035192	Nuclear Fuel I
89/07/11	Ifr265465	Shall
89/07/11	Isafr115:242	Shell
89/07/13	Iwo9101037	Schaffer

QUOTES FROM THE INTERNET

This column will present questions posted by actual skeptics of "cold fusion" to the c.f. bulletin board on the Internet (sci.physics.fusion). Also included are a few answers posted to the same net.

Q: In sci.physics.fusion [5357, 3/4/93] Dick Blue states:

== "As long as there are no ashes to be found,
== cold fusion is in doubt."

A: Ashes reported include tritium, charged particles, photon and phonon emission, and helium-4.

WHAT'S

Q: In sci.physics.fusion [5385; 3/6/93] Frank Close says:

== "You use the word censorship: do you mean that paper(s)
== submitted for publication have been rejected by the
== editor following peer review or do you mean something
== more sinister?"

HAPPENIN'

A: Frank, (many of us have submitted papers) but they are either accepted or rejected. Papers in this field are rejected because the field is said to "not exist". So since that is not "normal" peer review, what is left in your decision tree?

Q: In sci.physics.fusion [5403; 3/7/93; Matt Kennel writes:

== "How are you not sure that these are actually *created*, rather
== than being contaminations whose concentrations are enhanced by
== the reaction."

A: A good question, Matt. The careful investigators do that two ways. They measure the dynamic change over time. Also they do control experiments. That the changes, and control experiments, suggest de novo products leads thinking individuals to believe a scenario of nuclear fusion ought be considered.



NAGOYA, Japan 1992 --
THUMBS UP!! FOR
COLD FUSION

CF Historian Eugene Mallove (right) clearly makes a point to CF Archskeptics Douglas Morrison and Huizenga (left and center).

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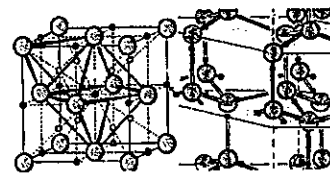
"In 1875 the director of the United States Patent Office sent in his resignation and advised that his department be closed. There was nothing left to invent, he claimed."

David Louis,
"2201 Fascinating FACTS",
Greenwich House,
NY (1983)

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TO COLDLY GO WHERE NO ONE
HAS GONE BEFORE

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Cold Fusion Slips Away

To the Editor:

"An Idealist's New Task: To Revamp Health Care," your otherwise excellent Feb. 26 profile of Ira C. Magaziner, President Clinton's senior adviser for policy development, does a great disservice to him in belittling his prophetic testimony to Congress on cold fusion in 1989.

Mr. Magaziner was right in 1989. As he warned and a Science Times article last Nov. 17 confirms, Japan has made the quest for cold fusion commercialization a priority, while support in the United States lags.

The obsolete Department of Energy negative report on cold fusion of 1989 doomed the United States to inaction, except for the Electric Power Research Institute and a handful of other companies. The Energy Department spends no money on cold fusion research, and the Patent Office — following Energy's line — refuses to grant patents to scores of inventors. Instead, for more than \$500 million a year the Energy Department pursues a doomed quest for commercial hot fusion power in 2050. Cold fusion applications will emerge this decade.

EUGENE F. MALLOVE
Bow, N.H., Feb. 26, 1993

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